Section 1.5 Sequences & summation

数列与求和

[AIME-2001-II-Q03] | Part A (Q1-Q5)

Answer: 898

1. Substituting the formula for x_{n-1} into that for x_n cancels all but $-x_{n-5}$, so $x_{n+5} = -x_n$ and $x_{n+10} = x_n$.

2. The indices reduce to $1, 3, 5$ modulo 10. Since $x_5 = 523 - 420 + 375 - 211 = 267$, the sum is $211 + 420 + 267 = 898$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-I-Q04] | Part A (Q1-Q5)

Answer: 840

1. Partial fractions give $a_k = 1/k - 1/(k+1)$, hence the sum is $1/m - 1/n = 1/29$.

2. Rearrange as $(29-m)(29+n) = 29^2$. Here $29-m > 0$ and $29+n > 29$, leaving only factors $1, 841$. Thus $m = 28, n = 812$, and the sum is 840.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2004-I-Q02] | Part A (Q1-Q5)

Answer: 201

1. The respective means are 2 and $1/2$. Thus m is odd, the maximum of A is $2 + (m-1)/2$, and that of B is m.

2. The condition is $|3-m|/2 = 99$. Its only positive solution is $m = 201$, which has the required odd parity.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q03] | Part A (Q1-Q5)

Answer: 802

1. For first term a and ratio r, $a/(1-r)=2005$ and $a^2/(1-r^2)=20050$. Dividing gives $a/(1+r)=10$.

2. Thus $2005(1-r)=10(1+r)$, so $r=1995/2015=399/403$. The terms are coprime and sum to 802.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q01] | Part A (Q1-Q5)

Answer: 100

1. Each four-term block $(4j)^2 + (4j-1)^2 - (4j-2)^2 - (4j-3)^2$ simplifies to $32j - 12$.

2. Sum for $j = 1, \ldots, 25$: $N = 32 \cdot 325 - 12 \cdot 25 = 10100$, remainder 100.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q03] | Part A (Q1-Q5)

Answer: 143

1. With first angle a and positive integer step d, the angle sum gives $18a + 153d = 2880$, or $2a + 17d = 320$.

2. Parity forces even $d \geq 2$. Convexity requires $a + 17d = (320 + 17d)/2 < 180$, so $17d < 40$.

3. Therefore $d = 2$ and $a = 143$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q05] | Part A (Q1-Q5)

Answer: 542

1. Group terms into blocks of 2011. Their sums form a geometric sequence, and its first two terms are $200, 180$.

2. The block ratio is $9/10$, so the third block sums to 162. The requested total is $200 + 180 + 162 = 542$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2012-I-Q02] | Part A (Q1-Q5)

Answer: 195

1. If there are n terms, the total increase is the sum of the first n odd numbers, $n^2 = 121$. Thus $n = 11$.

2. The middle term equals the mean, $715/11 = 65$, and the two endpoint terms sum to twice this. The requested sum is 195.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2012-II-Q02] | Part A (Q1-Q5)

Answer: 363

1. For common ratio $r \neq 0$, $27r^{14} = 99r^{10}$ gives $r^4 = 11/3$.

2. Therefore $a_9 = 27r^8 = 27(11/3)^2 = 363$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2013-II-Q03] | Part A (Q1-Q5)

Answer: 350

1. Burning j complete centimeters takes $10\sum_{k=1}^{j} k = 5j(j+1)$ seconds. Thus $T/2 = (5 \cdot 119 \cdot 120)/2 = 35700$.

2. The equation $j(j+1) = 7140 = 84 \cdot 85$ shows exactly eighty-four centimeters have burned. Thus $h = 35$ and $10h = 350$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q01] | Part A (Q1-Q5)

Answer: 722

1. Pair terms with the common factor $2k$. Their difference in $B - A$ is $2k[(2k+1) - (2k-1)] = 4k$.

2. Therefore $B - A = 1 - 39 + 4\sum_{k=1}^{19} k = -38 + 4 \cdot 190 = 722$. This is positive, so its absolute value is 722.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q01] | Part A (Q1-Q5)

Answer: 336

1. Geometric summation gives $S(a)S(-a) = 144/(1-a^2) = 2016$, so $(1-a^2)^{-1} = 14$.

2. The sum is $12/(1-a) + 12/(1+a) = 24/(1-a^2) = 336$. The resulting $a^2 = 13/14$ is consistent with convergence.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q05] | Part A (Q1-Q5)

Answer: 053

1. If there are $k \geq 2$ days, arithmetic summation yields $k(2n + k - 1) = 748$ and $k(2t + k - 1) = 638$.

2. Subtraction gives $k(n - t) = 55$. Also k divides both $748, 638$, so it divides 22. Thus $k \mid \gcd(55, 22) = 11$, forcing $k = 11$.

3. Now $2n + 10 = 68$ and $2t + 10 = 58$, giving $n = 29, t = 24$. Hence $n + t = 53$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q01] | Part A (Q1-Q5)

Answer: 108

1. The remaining total is 405, so the middle arithmetic term is 135. Thus $B = 144$ initially and $A + C = 300$.

2. Geometric progression gives $AC = B^2 = 20736$. The roots of $t^2 - 300t + 20736 = 0$ are $108, 192$. Therefore $A = 108$, and the remaining counts $103, 135, 167$ check the arithmetic condition.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2023-II-Q01] | Part A (Q1-Q5)

Answer: 220

1. Let the least count be a and greatest be $2a$. The sum of six terms is $6(a+2a)/2 = 9a$.

2. Thus $a = 990/9 = 110$, and the largest count is 220.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2026-II-Q01] | Part A (Q1-Q5)

Answer: 178

1. The common difference must be a positive divisor of both 20 and 30, hence one of $1, 2, 5, 10$. Each such divisor works.

2. The tenth term is $4 + 9d$, so the sum is $4 \cdot 4 + 9(1 + 2 + 5 + 10) = 178$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q06] | Part B (Q6-Q10)

Answer: 521

1. Use $1/(n^2-4) = \frac{1}{4}(1/(n-2) - 1/(n+2))$. The value becomes $250(1 + 1/2 + 1/3 + 1/4) - R = 520 + 5/6 - R$, where $R = 250(1/9999 + 1/10000 + 1/10001 + 1/10002)$.

2. The bound $0 < R < 1000/9999 < 1/3$ puts the value strictly between 520.5 and $520 + 5/6$. Thus the nearest integer is 521.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2003-II-Q08] | Part B (Q6-Q10)

Answer: 348

1. Index from $j = 0$. Its term formula is quadratic, $T_j = aj^2 + bj + c$. The first values give $c = 1440, a + b = 276, 4a + 2b = 408$.

2. Thus $a = -72, b = 348$. The eighth term is $T_7 = -72 \cdot 49 + 348 \cdot 7 + 1440 = 348$.

3. The factorization $T_j = (180 - 24j)(8 + 3j)$ confirms that two real arithmetic sequences realize the formula.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2004-II-Q09] | Part B (Q6-Q10)

Answer: 973

1. Put $a_2 = r$. Induction gives $a_{2k+1} = [1 + k(r-1)]^2$ and $a_{2k+2} = [1 + k(r-1)][1 + (k+1)(r-1)]$.

2. The given sum becomes $(4r-3)(9r-7) = 646$, or $(r-5)(36r+125) = 0$. Positivity gives $r = 5$.

3. Thus the terms are $(4k+1)^2$ and $(4k+1)(4k+5)$, increasing with the index. The boundary terms are $a_{16} = 29 \cdot 33 = 957$, $a_{17} = 1089$. The requested sum is $16 + 957 = 973$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-I-Q06] | Part B (Q6-Q10)

Answer: 017

1. Induction gives row-r, position-j entry $2^{r-1}(r+2j-2)$, for $1 \leq j \leq 51-r$.

2. As sixty-seven is odd, divisibility requires $r+2j-2=67$, the only positive multiple of sixty-seven within the possible range up to ninety-nine. It occurs exactly for odd $r \leq 33$.

3. There are seventeen such rows, each with one qualifying entry.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q06] | Part B (Q6-Q10)

Answer: 561

1. Dividing by u_{n-1} shows successive ratios increase by one: $u_n/u_{n-1} = 1 + u_{n-1}/u_{n-2}$.

2. Therefore $a_n = n!$, $b_n = (n+2)!/2$. Their ratio at thirty-two is $34 \cdot 33/2 = 561$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q07] | Part B (Q6-Q10)

Answer: 041

1. Rewrite as $5^{a_{n+1}-a_n} = (3n+5)/(3n+2)$. Multiplication telescopes to $5^{a_k-1} = (3k+2)/5$, hence $a_k = \log_5(3k+2)$.

2. Thus $3k+2 = 5^j$. Modulo three this requires odd j; $j=1$ gives excluded $k=1$, and the next is $j=3$, giving $k=41$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q08] | Part B (Q6-Q10)

Answer: 398

1. Differences are all distinct: $2^j - 2^i = 2^i(2^{j-i} - 1)$ has two-adic valuation i, then the odd part uniquely determines j. Thus summing pairs counts each difference once.

2. The coefficient of 2^j is $j - (10 - j) = 2j - 10$. Hence $N = 2\sum_{j=0}^{10} j2^j - 10\sum_{j=0}^{10} 2^j = 2(18434) - 10(2047) = 16398$.

3. The remainder is 398.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2014-II-Q07] | Part B (Q6-Q10)

Answer: 021

1. At integer k, $f(k) = [(k+1)(k+2)]^{(-1)^k}$. The sum of logs is the log of their telescoping product.

2. For even n, the product is $(n+2)/2$; for odd n, it is $1/[2(n+2)]$. Absolute log one means a product of ten or one-tenth.

3. The even case gives $n = 18$, and the odd case $n = 3$. No other sign choices fit their product ranges. Their sum is 21.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q09] | Part B (Q6-Q10)

Answer: 494

1. An adjacent equal pair forces a zero two steps later. An adjacent pair differing by 1 first creates an equal pair, then a zero. Thus all triples with $|a_2 - a_1| \leq 1$ or $|a_3 - a_2| \leq 1$ qualify.

2. Count the complement by fixing a_2. At either endpoint $1, 10$, each neighbor has 8 choices at distance at least 2; at each other value there are 7 choices. Hence $2 \cdot 8^2 + 8 \cdot 7^2 = 520$ avoid these initial triggers, leaving 480 qualifying triples.

3. Among the 520, if $a_3 \geq 2$, then $a_4 \geq 2a_3$ and the subsequent terms increase with every adjacent gap at least 2, so zero never occurs. If $a_3 = 1$, the only additional trigger is $|a_2 - a_1| = 2$, making $a_4 = 2$ and an adjacent gap 1.

4. For $a_3 = 1$, the previous gap restriction requires $a_2 \geq 3$. Pairs $a_1 = a_2 - 2$ give 8 choices, and $a_1 = a_2 + 2$ give 6. Larger first gaps produce $a_4 \geq 3$ and permanent growth. Therefore the total is $480 + 14 = 494$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q10] | Part B (Q6-Q10)

Answer: 504

1. Write the initial reduced ratio as $b/a > 1$. Integrality gives the first three terms ca^2, cab, cb^2 for a positive integer c. Put $d = b - a \geq 1$. Inducting through the two triple rules yields $a_{2j+1} = c(a + jd)^2$ and $a_{2j+2} = c(a + jd)(a + (j + 1)d)$ for $j \geq 0$.

2. Thus $c(a + 6d)^2 = 2016 = 2^5 \cdot 3^2 \cdot 7$. The integer $N = a + 6d \geq 7$ has square dividing2016, so it divides12. The only divisor of12 at least7 is12.

3. Hence $c = 14$ and $a + 6d = 12$. Positivity forces $d = 1, a = 6$, with $b = 7$ coprime to a. The displayed formulas give an increasing integral sequence for all indices, verifying existence.

4. The requested first term is $14 \cdot 6^2 = 504$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q09] | Part B (Q6-Q10)

Answer: 262

1. Write $a_n = 1 + (n-1)d$ and $b_n = r^{n-1}$, where $d \geq 1, r \geq 2$ are integers. Since $c_1 = 2$, $k \geq 3$. The conditions become $(k-2)d + r^{k-2} = 99$ and $kd + r^k = 999$.

2. Subtracting gives $2d + r^{k-2}(r^2 - 1) = 900$. The power term is divisible by3, hence $3 \mid d$. The first equation then forces $3 \mid r$.

3. Bounds $r^{k-2} < 99, r^k < 999$ leave $(r,k) = (3,3), (3,4), (3,5), (3,6), (6,3), (9,3)$. Substituting into the two equations, only $(9,3)$ works, with $d = 90$.

4. Thus $c_3 = 1 + 2 \cdot 90 + 9^2 = 262$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2020-II-Q06] | Part B (Q6-Q10)

Answer: 626

1. Rescale by $u_n = 5t_n$. Then $u_n = (u_{n-1} + 1)/u_{n-2}$. Starting from arbitrary positive a, b, the next three values are $(b+1)/a$, $(a+b+1)/(ab)$, $(a+1)/b$, followed by a, b.

2. The recurrence therefore repeats every five indices. Since $2020 \equiv 0 \pmod 5$, use $u_5 = (100+1)/105 = 101/105$.

3. It follows that $t_{2020} = 101/525$ in lowest terms, and the answer is 626.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q11] | Part C (Q11-Q15)

Answer: 518

1. A sum-one geometric series with ratio r begins with $1-r$, so its second term is $r(1-r)$. Distinct ratios giving the same value must be $r, 1-r$.

2. The third-term condition gives $r^2(1-r) = 1/8$. With $t = 2r$, this is $(t-1)(t^2-t-1) = 0$. The root $r = 1/2$ gives identical series; the negative root makes the other ratio exceed one. Thus $r = (1+\sqrt{5})/4$.

3. The common second term is $(\sqrt{5}-1)/8$, so the requested encoding is $500 + 10 + 8 = 518$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q11] | Part C (Q11-Q15)

Answer: 889

1. Multiply by a_k. The adjacent products $b_k = a_k a_{k-1}$ satisfy $b_{k+1} = b_k - 3$, with $b_1 = 2664$. Hence $b_k = 3(889 - k)$.

2. For $k \leq 888$ these products are positive, so all needed denominators remain nonzero. At $k = 889$ the product vanishes and $a_{888} \neq 0$, forcing the first zero $a_{889} = 0$. Thus $m = 889$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2006-I-Q15] | Part C (Q11-Q15)

Answer: 027

1. Squaring gives $x_k^2 - x_{k-1}^2 = 6x_{k-1} + 9$. Summing through $k = 2007$ yields $\sum_{k=1}^{2006} x_k = (x_{2007}^2 - 18063)/6$.

2. Induction shows $x_k/3$ is an integer of parity k. Thus x_{2007} is an odd multiple of three. The nearest such square to 18063 is $135^2 = 18225$; neighboring candidates $129^2, 141^2$ are farther. The resulting absolute sum is 27.

3. Attain it by $x_k = 3k$ through $k = 45$, then alternate -138 at even indices and 135 at odd indices. This obeys the recurrence and finishes at $x_{2007} = 135$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2006-II-Q11] | Part C (Q11-Q15)

Answer: 834

1. For $S_n = \sum_{j=1}^{n} a_j$, induction gives $2S_n = a_{n+2} + a_n$. It starts at $n = 1$, and adding $2a_{n+1}$ uses exactly the tribonacci recurrence.

2. Thus $S_{28} = (20603361 + 6090307)/2 = 13346834$, remainder 834.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2007-I-Q14] | Part C (Q11-Q15)

Answer: 224

1. The invariant $(a_{n+1}+a_{n-1})/a_n$ is constant: subtract consecutive product recurrences and factor to equate its consecutive values. Since $a_3 = 672$, that constant is $(672+3)/3 = 225$.

2. Thus $a_{n+1} = 225a_n - a_{n-1}$ and $a_{n+1}^2 + a_n^2 + 2007 = 225a_n a_{n+1}$. Positive terms increase strictly after a_2, so the late adjacent product exceeds 2007.

3. The target ratio is $225 - 2007/(a_{2006}a_{2007})$, strictly between 224 and 225. Its floor is 224.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2007-II-Q12] | Part C (Q11-Q15)

Answer: 091

1. Write $x_j = 3^{u+vj}$ with integer $v \geq 1$. The logarithm sum gives $2u + 7v = 77$.

2. The positive sum lies strictly between its largest term x_7 and $3x_7$, since its ratio backwards is at most one third. Its logarithm therefore lies strictly between integers $u + 7v$ and $u + 7v + 1$, forcing $u + 7v = 56$.

3. Solve $u = 21, v = 5$, giving $u + 14v = 91$.

Source / 来源

Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q11] | Part C (Q11-Q15)

Answer: 073

1. Expand the determinant along its first row, then the first column of the off-diagonal minor. This gives $D_n = 10D_{n-1} - 9D_{n-2}$, using $D_0 = 1, D_1 = 10$.

2. The recurrence has characteristic roots $9, 1$. Its initial values give $D_n = (9^{n+1} - 1)/8$, also verifiable directly by induction.

3. Therefore the sum is $\sum_{n=1}^{\infty} 9^{-(n+1)} = (1/81)/(1 - 1/9) = 1/72$. It converges geometrically, and $p + q = 73$.

Source / 来源